

# **Computer Security - CheatSheet**

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**CompSec Properties - Confidentiality.** prevent unauthorized disclosure of information. *authorized users may read a file*

- **Integrity.** prevent unauthorized modification of information. *authorized programs may write a file*

- **Availability.** prevent unauthorized denial of service or access to information and resources. *authorized services can access a file*

- **Authenticity.** assurance that entities (users, systems, or data) are genuine and can be verified as such.

- **Anonymity.** protection of an individual's identity from being disclosed or linked to specific actions or data.

- **Non-repudiation.** assurance that a party in a communication cannot deny the authenticity of their signature or the sending of a message.

**The Adversary.** malicious entity aiming at breaching the security policy and **will** choose the optimal way to use its ressources to mount an attack that violates the security properties.

**Threat Model.** describes the ressources available to the adversary and their capabilities *(has access to internet, but doesn't have access to the internal network of the company.)*

**Threat.** Who might attack which assets, using what resources, with what goal, how, and with what probability

**Vulnerability.** Specific weakness that adversaries could exploit with interest in a lot of assets *(API is not protected, password appears in plain text...)*

Principles of CompSec.

- 1. **Economy of mechanism** Keep security mechanism design simple and small ⇒ Easier to audit and verify, testing is **not** appropriate to evaluate security.
- **Trusted Computing Base (TCB).** Every component of the system on which the security policy relies upon *hardware, firmware, software.*  
The TCB is trusted to operate correctly for the security policy to hold. → If something goes wrong in it, the security policy may be violated
- must** be kept small to easy verification / diminish the attack surface
- 2. **Fail-safe defaults.** Base access decisions on permission rather than exclusion. *(Whitelist, do not blacklist)*  
If something fails, be as secure as it does not fail errors / uncertainty should error on the side of the security policy.
- 3. **Complete mediation..** Every access to every object must be checked for authority. A **Reference Monitor** mediates all actions from subjects on objects and ensures they are according to the policy. *△time to check vs. time to use*
- 4. **Open design** The design should not be secret *Always design as if the enemy knows the system. (Crypto-only secret is key.Authentication-Only secret is password.for Obfuscation-Only used secret is noise.).assuming the thread model can't get a hold of the system is unrealistic (employee corruption, ...)*

**Access Control.** Security mechanism that ensures all accesses and actions on objects by principals are within the security policy.

**no chicken soup** *(checks everywhere in code), use a reference monitor, module used all over code checking for subjects and actions*

**Discretionary Access Control (DAC).** Object owners assign permissions, ownership of resources. *Linux, Social Networks*

**Mandatory Access Control (MAC).** Central security policy assigns permissions, usually for organizations with need for central control. *Military, Hospital, ...*

**Access Control Matrix.** abstract representation of all permitted triplets of (subject, object, access right) within a system.

|       | file1       | file2       | file3       |
|-------|-------------|-------------|-------------|
| Alice | read, write |             | read        |
| Bob   |             | read, write | read, write |

*Complexity. O(f · u)*

**Access Control List (ACL).** associate permissions to objects, stores permissions close to the resource.

*file1: (Alice,read/write), file2: (Bob, read/write), file3: (Alice,read),(Bob,read/write)*

⊗ easy to determine who can access a resource and to revoke rights by resource.  
⊗ hard to check all users rights, to remove all perms. for a user, delegate perms.

**Role Based Access Control (RBAC).** access granted based on user roles and predefined permissions. systems have too many subjects (that come and go) → large dynamic ACLs. Subjects are often similar to each other and get assigned the same rights.

1. Assign permissions to roles, 2. Assign roles to subjects

3. Subjects have the permissions of their assigned role

⊗ role explosion (temptation to create fine grained roles), limited expressiveness, difficult to implement separation of privilege + least privilege.

**Group Based Access Control (GBAC).** access granted based on user group membership. **exactly like RBAC but instead of roles, permissions are assigned to groups.** groups are broader, represent organizational units instead than specific functions.

1. Assign permissions to groups 2. Assign subjects to groups

3. Subjects have the combined permissions of all their groups

⊗ coarse granularity, overlapping group memberships, inconsistent permissions, difficult to manage if users belong to many groups.

*In case of Negative Permissions check negative permissions first before group, △system crashes before negative checks.*

**Ambient Authority.** an action succeeds if the subject only specifies the *operation* and the *object name*, not the specific authority used.

⊗ leads to accidental misuse of authority, programs may act with more rights than intended. → Confused Deputy Problem.

**Confused Deputy Problem.** when a program (deputy) is tricked into misusing its authority on behalf of another subject.

*Alice runs compiler(input, bill) ⇒ compiler (with write access to bill) overwrites billing file. ⇒ Alice uses compiler's authority to modify bill indirectly.*

⊗ ambient authority allows unintended privilege use.

**Solutions:**

1. Restrict privileged process access.
2. Make privileged process check user's authorization.
3. Use **Capabilities** to explicitly delegate rights.

**Harm.** The bad thing that could happen when the **threat** materialsizes. *(adversary steals the money, learns my password...)*

**Security Policy.** high level description of the security properties that must hold in the system in relation to assets and principals.

- **Assets (objects).** anything with value (data, files, memory) needing protection.

- **Principals (subjects).** people, computer programs, services.

**Security Mechanism.** Technical mechanism used to ensure that the security policy is not violated by an adversary within the threat model, **we can only prepare for threats we're aware of**

*(Policy. ensure messages cannot be read by anyone but the sender and the receiver, Mechanism. encrypt the message before sending)*

**Composition of Security Mechanisms**

- **Defence in depth.** As long as one remains unbroken the Security Policy isn't broken) *(two-factor auth)*
- **Weakest Link.** if anyone fails the Security Policy, it is broken. *(security questions for a lost password → just need to know the answer...)*

*Humans can be vulnerabilities - phishing attacks, bad use of passwords...)*

**To show a system is secure. (under a specific threat model)**

- Attacker - Just one way to violate **one** security property is enough.
- Defender - No adversary strategy can violate the security policy.

**Security Argument.** Rigorous argument that security mechanisms in place are effective in maintaining security policy subject to assumptions on Threat Model.

- 5. **Separation of privilege.** No single accident, deception, or breach of trust is sufficient to compromise the protected information
- **Privilege.** A privilege allows a user to perform an action on a computer system that may have security consequences. *(create a file in a directory...)*
- 6. **Least Privilege.** Every program and every user of the system should operate using the least set of privileges necessary to complete the job. *Rights are added on need, discarded after use. Users should only know about things if they have to.*
- 7. **Least Common Mechanism** Minimize the amount of mechanism common to more than one user and depended on by all users. Every shared mechanism represents a potential information path between users.
- 8. **Psychological acceptability.** It is essential that the human interface be designed for ease of use, so that users routinely and automatically apply the protection mechanisms correctly. *(hide complexity, cultural acceptability...)*
- 9. **Work Factor**  
Compare the cost of breaking the mechanism with the resources of a potential attacker. *(cost of compromising insiders, cost of finding a bug, monetization...)*
- 10. **Compromise recording** Detect and record security breaches with tamper-evident logs, ensuring traceability, integrity, confidentiality, and availability of recorded data.

**User & Group Identities.** Most modern systems rely on DAC.

**UIDs / GIDs.** numerical identifiers for users and groups.

*/etc/passwd:* username:password:UID:GID:info:home:shell

*/etc/group:* defines secondary groups.

Each user has a home directory and belongs to one or more groups.

**UNIX Model.** Everything is a file.

| directories | owner      | group | others   | owner    | group |               |            |  |  |  |  |  |  |  |  |  |  |  |  |
|-------------|------------|-------|----------|----------|-------|---------------|------------|--|--|--|--|--|--|--|--|--|--|--|--|
|             | drwxrwxr-x | 1     | catronco | catronco | 4096  | Sep 16 14:23  | exampledir |  |  |  |  |  |  |  |  |  |  |  |  |
|             | -rwxrwxr-w | 1     | catronco | catronco | 8600  | Sep 15 15:20  | hello      |  |  |  |  |  |  |  |  |  |  |  |  |
|             | -rw-rw-rw- | 1     | catronco | catronco | 150   | Sep 15 15:14  | hello.c    |  |  |  |  |  |  |  |  |  |  |  |  |
| files       | -rw-rw--w- | 1     | catronco | catronco | 45    | Sep 15 15:07  | test1.txt  |  |  |  |  |  |  |  |  |  |  |  |  |
|             |            | links |          |          | size  | last modified | filename   |  |  |  |  |  |  |  |  |  |  |  |  |

**Directories.** Read → list files; Write → add/remove files; Exec → traverse (cd).

**Permission check order:** 1. If process UID = file owner → check owner bits.  
2. Else if GID matches → check group bits.  
3. Else → check “other” bits.

*Root (UID 0. bypasses all checks.*

**Changing Permissions.**

**chmod.** modify permission bits. *chmod +r file, chmod 666 file.*

**chown.** change file owner/group(opt.) *chown root:staff /srv/config*

**chgrp.** change file group. *chgrp www-data /var/www*

**Special Rights.**

**sudo (set user ID).** run program with owner's privileges. *puts s in owner's x field*  
*chmod u+s filename*  
*allows normal users to change passwords without full root access.*

**sgid (set group ID).** run program with group's privileges.  
*on a directory - creating a file inside gets folder's group*  
*d rwx r-s r-x 2 root staff /srv/shared*

**sticky bit.** on directories, prevents deleting/rename files you don't own.  
*d rwx rwx rwt 10 root tmp /tmp*

**Special Users.**

*root:* UID 0, full privileges, bypasses checks, in TCB.

*sudo run as root su [username] switch to account(default is root)*

*nobody:* UID -2, owns no files, minimal privileges, safer for untrusted code.(least privilege)

⊗ flexible, simple model, widely adopted.

⊗ relies on **ambient authority**, prone to Confused Deputy attacks.

**Windows: DACL.** Controls access to objects via list of ACEs.

**ACE (Access Control Entry).** <Type, Principal, Permissions, Flags>.

- **Types.** Allow / Deny (negative / positive). Deny takes precedence and ordered with Denied permissions first.
- **Principal.** User or group (SID).
- **Permissions.** Fine-grained rights(*Read, Write, Execute, Delete...*)
- **Flags.** Inheritance, propagation, object-specific.

**Access Tokens.** Thread/process carries user + group SIDs checked against DACL.

**Least Privilege.** Users run limited by default; elevation via “Run as admin.”

**Capabilities.** associate permissions to *subjects*

*Alice: {(file1, read/write)} Bob: {(file2, read/write), (file3, read/write)}*

⊗ easy to determine all permissions of a user and to delegate rights by subject.

⊗ hard to determine who can access a resource or to revoke rights by resource.

**Model.** a design pattern for a set of properties. (△ not covered by model - who are the subjects? what are the objects? what mechanisms to use to implement it?)

**Bell-La Padula (BLP).**  
security model where Subjects S and objects O are associated to a level of **Confidentiality**.  
**Access rights.** Execute, Read, Append, Write.  
- Objects are associated to a Security Level = (Classification, set of categories).  
*{{Unclassified < Confidentiality < Secret < Top Secret}, {NATO, Crypto, Nuclear}}*  
**Dominance Relationship.**  
Transitive, There always is a Top and Bottom, Only partial ordering (some pairs of elements can't be compared).  
A security level ( $l_1, c_1$ . dominates ( $l_2, c_2$ . if and only if  $l_2 \leq l_1$  and  $c_2 \leq c_1$ .  
*eg. (Secret, {Nuclear, Army}) ≥ (Confidential, {Army})*  
**Clearance.** max security level a subject has been assigned. *clearance-level( $S_i$ )*.  
**Current Security level.** subjects can operate at lower security levels. *current-level( $S_i$ )*.  
**Declassification.** Removing classification labels from information so it can be shared with lower security levels. *soldier (S) with Secret clearance shares sanitized info like "We've seen tanks moving" with a lower-level user (U)*.

**Covert Channels.** Any channel that allows information flows contrary to the security policy, *Storage channels(shared counters, ...), Timing channels(queueing time...)*.  
△Least Common Mechanism. *mitigated by adding noise or isolation(no high ↔ low level communication)*.

**BIBA (Integrity).**  
Security model where subjects S and objects O are associated to a level of **Integrity** (trustworthiness).  
**Access rights.** Read (Observe), Write (Modify)  
**BIBA to create Integrity Policies.**  
**Simple Integrity Property.**  
If ( $s, o, r$ ) is a current access, then level(s) dominates level(o).  
**(SUBJECTS CAN'T READ DOWN)** ⇒ prevents contamination from low to high.  
**\*-Integrity Property.**  
If ( $s, o, w$ ) is a current access, then level(o) *does not* dominate level(s).  
**(SUBJECTS CAN'T WRITE UP)** ⇒ prevents low from corrupting high state.  
**Discretionary Property (DAC).** Subject must still have explicit permission for the access (r, w, x) per the access control matrix.  
**Basic Integrity Theorem (induction).** If the initial state is integrity-secure and all state transitions satisfy BIBA properties, then all reachable states preserve integrity.

**Sanitization (lifting low → high).**  
- *Fail-safe default:* deny by default; elevate only after checks pass.  
- *Whitelist over blacklist:* validate that all required properties of "good" hold.  
- *Context-aware:* encoding, schema, range, semantics, and provenance checks.  
*Note:* Sanitization bugs commonly break integrity guarantees.  
**Principles for Integrity.**  
- **Separation of Duties.** Require multiple principals to perform an operation  
- **Rotation of Duties.** Allow a principal only a limited time on any particular role and limit other actions while in this role  
- **Secure Logging.** Tamper evident log to recover from integrity failures. Consistency of log across multiple entities is key.

**Chinese Wall Model.** All objects are associated with a label denoting their origin.  
*(Pepsi, Coca-Cola, Microsoft Audit, Microsoft Investments)*  
Define **conflict sets** of labels. *{Pepsi, Coca-Cola}, {Microsoft Audit, Microsoft Investments}*.  
Subjects are associated with a **history** of their accesses to objects and their labels.  
**Access Rules.** A subject can access an object (read or write) **only if** the access does not allow information flow between items with labels in the same conflict set.  
**Example (Direct Flow).**  
1. Access to Pepsi (OK)  
2. Access to Microsoft Invest (OK)  
3. Access to Coca-Cola (Denied)  
**Example (Indirect Flow).**  
1. Alice accesses Pepsi  
2. Bob accesses Coca-Cola and IBM  
3. Alice tries to access IBM (*Denied, indirect link via Bob*)  
**Sanitization.**  
Allows more flexibility by "un-labeling" some items→controlled sharing/reuse of data.

**All together (Asymmetric Cryptography) - Bob sends a message to Alice**  
**1. Get public keys (via PKI).**  
Bob retrieves Alice's encryption public key  $PK_{\text{Alice}}$  and verification public key  $PK_{\text{Alice}}$ .  
**2. Prepare the message.**  
Bob has message  $M$  and computes its hash  $h(M)$ .  
Allows: the signature to cover the exact content of  $M$  while keeping computation efficient.  
**3. Sign the message hash.**  
Bob uses his secret signing key  $SK_{\text{Bob}}$  to generate a digital signature.  $\text{Sign}_{SK_{\text{Bob}}}(h(M))$ .  
**4. Encrypt the message.**  
Bob encrypts the message using Alice's encryption public key  $PK_{\text{Alice}}$ :  $E_{PK_{\text{Alice}}}(M)$ .  
Allows: only Alice, who holds the matching secret key  $SK_{\text{Alice}}$ , to read the message.  
**Authenticity.** only sender can generate a valid signature, allowing the server to verify the sender's identity and preventing impersonation. **Integrity.** Any message change invalidates signature, ensuring tampering is detectable. **Non-repudiation.** only sender can produce the valid signature → they can't deny later having sent the message.

**Block Cipher - Electronic Code Block (ECB).**  
encrypt each message block independently using the same key.  
1. The message  $M$  is divided into blocks.  
 $M = M_1 M_2 M_3 M_4$ .  
2. Each block is encrypted separately  $C_i = E_K(M_i)$ .  
3. Decryption.  $M_i = D_K(C_i)$ .  
△If ( $M_i = M_j$ ) ⇒ ( $C_i = C_j$ ) ⇒ (freq. analysis)

**BLP to create Confidentiality Policies.**  
**Simple Security Property.**  
if (subject, object, w/r) is a current access, ⇒ level(subject) dominates level(object).  
**(SUBJECTS CAN'T READ UP)**  
**Star Property.** If a subject has simultaneous "observe" (r,w) access  $O_1$  and "alter" ( $a, w$ ) access to  $O_2$  then level  $O_2$  dominates level  $O_1$ .  
**(SUBJECTS CAN'T WRITE DOWN).** *changing object's perms is write-like.*  
**Discretionary Property.** A subject may only access an object if it has explicit permission for that specific type of access (r, w, x, etc.) defined by the access control matrix.

**Basic Security Theorem(induction).** if all state transitions are secure, and the initial state is secure, then every subsequent state is secure regardless of the input.

**BIBA Low-Water-Mark Variants (taint-tracking).**  
**For Subjects.**  
Subjects start at their max integrity, **on read:**  $\text{current-level}(s) = \min(\text{current-level}(s), \text{level}(o))$ .  
⇒ once tainted, subject sinks; prevents writing up thereafter.  
**For Objects. On write by s:**  
 $\text{level}(o) = \min(\text{level}(o), \text{level}(s))$ .  
⇒ objects "sink" easily; can cause integrity collapse.  
*Mitigation:* replicate high/low objects; sanitize before promotion; detect/flag unexpected level drops.  
**BIBA Additional Actions - Invoke.**  
**Simple Invocation.** only allow subjects to invoke subjects with a label they dominate.  
⊕ protect high integrity data from misues by low integrity principals  
⊖ what level is the output?  
**Controlled Invocation.** Only allow subjects to invoke subjects that dominate them.  
⊕ prevents corruption of high integrity data  
⊖ hard to detect polluting information.

**Cryptography.**  
**Data in transit.** Securing communications. **Data at rest.** Securing stored informations  
Let C the Ciphertext, K the Key, M the Message/Plaintext.  $C = E_K(M)$  and  $M = D_K(C)$   
**Keyspace.** Number of possible keys that can be used in an encryption algorithm.  
**Invertibility Requirement.**  $\forall K, M, D_K(E_K(M)) = M$  (otherwise can't recover message)  
**Security Requirement.** Functions should be hard to invert without knowing K.  
Ideal Case - adversary must try every possible combination of keys (bruteforce).  
*Caesar's Cipher*  
- Encryption. Shift each letter by a fixed number (K)  
- Decryption. Shift each letter by a fixed number (-K)  
Keyspace. Only 25 possible keys.  
 $\log_2(25) = 4.6$  bits of security (too small).  
Both can be broken using **Frequency Analysis Attack**.

**Frequency Analysis.** Use statistical properties of the language.  
1. Most frequent letter in English is 'e'. 2. Identify most frequent letter in Ciphertext. 3. Map it to 'e'.  
**Ideally,** An N-bit key should offer security as close to N bits as possible (require  $2^N$  attempts), if not the algorithm is considered **broken**.  
**Types of Adversaries.** security models  
- Passive Eavesdropper - adversary can only read the ciphertext  
- Active Attacker - adversary can influence the system (*corruption of one of parties, ...*)  
**Known Plaintext Attack (KPA).** **Chosen Plaintext Attack (CPA).**  
Active security model. Attacker gets - Suppose Eve convinces Alice to encrypt chosen messages access to some pairs, (message, cipher- with the secret key text), all encrypted with secret K. - For all  $m_i$  Eve gets access to  $E_K(m_i)$ . Eve has access to an **Encryption Oracle**.  
**From these pairs, she tries to guess key** crypton Oracle.  
**K to decrypt other messages** (*realistic, she could get access to an encrypted messaging app*  
*realistic because of message headers, for limited time, or to an encryption api*).  
message headers ("From:...", times-△ if Eve encrypts entire alphabet she can reveal the key instantly).

**Side-Channels Attacks**  
- Timing attack - Eve measure how long it takes for a given message to get encrypted or a ciphertext to be decrypted  
- Power Analysis - Eve observes the energy consumed by the device doing the crypto.  
**One-Time-Pad (OTP).** OTP is **Perfectly Secure**.  
*Goal - Remove frequency analysis*  
- Use a key of random bits as long as the message.  
-  $\text{Enc}(k, m) = m \oplus k$   
-  $\text{Dec}(k, m) = \text{Enc}(k, m) \oplus k$   
Key should be random for every sent message.  $\forall m, c, P(M = m | E_K(m) = c) = P(M = m)$ .  
Otherwise attacker can collect information about the message.  
*Integrity Attack Example (Eve flips the first bit of M):*  
1. Eve flips the first of C to get C'  
2. Bob decrypts C':  $M' = C' \oplus K = (M \oplus K \oplus \Delta) \oplus K = M \oplus \Delta$   
**Symmetric Cryptographic Schemes** **Symmetric Cryptographic key**  
Encryption/decryption done with the **same key** **- Known to both parties.** Partners must agree on the key before starting using the primitive  
- **Block Ciphers.** Operate on fixed-size blocks.  
- **Stream Ciphers.** Operate on bit/byte at a time, like pseudo-OTP.  
- **Reused.** The keys is pre-shared once and then reused (*but keys have a "duration"*)  
- **Must be secret.** Revealing the key eliminates any protection provided by the primitive

**Stream Ciphers - The pseudo-OTP** emulate OTP while solving key-length problem.  
1. Secret short Key (K) shared between Alice and Bob 2. Key Stream Generator - Uses K and Initialization Vector (IV) → an arbitrary long, pseudo-random bit stream (S). 3. Encryption.  $C = M \oplus S$  (*generator needs a key as main seed to generate a predictable random sequence, an iv for the sequence to start differently on each run*)  
⊕ Speed, Low Error propagation(errors in one bit do not affect subsequent symbols)  
⊖ Low Diffusion (a change in a bit only affects one bit), Susceptibility to Modification (low diffusion makes it easier to tamper)  
⊖ key stream generators are periodic because seed is finite. Thus, we need period long enough to not an issue (avoiding frequency analysis)

**Linear Feedback Shift Register for Key Stream Generators.** build a "random" sequence of bits using a linear recurrence relation on a sequence of bit.  
*Example: Starting with state  $a_0, a_1, a_2, a_3 | a_n = a_{n-3} \oplus a_{n-4}$  (⊕ Easy to build/analyze.)*  
**Randomness of LFSR.** **Distribution property.**  
if characteristic polynomial of the recurrence relation of the LFSR is primitive. The maximum possible number of states before repeating. For an L-bit register, the maximum period is  $2^L - 1$  states  
As a consequence, the sequence generated exhibit good distribution properties. Every possible non-zero state appears exactly once in a cycle.  
⊖ underlying operation is **linear**. (*some algorithms can recover this with a stream subsequence*)

Symmetric ciphers are fast, but require a **pre-shared secret key**  
**Diffie-Hellman Setup.** Alice, Bob (and Eve) know large public parameters: prime p/generator g  
*Alice's Actions* *Bob's Actions*  
1. Chooses **private** secret a. 1. Chooses **private** secret b.  
2. Computes Public Value  $A = g^a \text{ mod } p$  2. Computes Public Value  $B = g^b \text{ mod } p$   
3. Sends A to Bob. 3. Sends B to Alice  
Eve sees A, B, g, and p, but can't find a/b due to the **Discrete Log Problem**.  
Both Alice and Bob can compute a shared secret K:  $K = B^a \text{ mod } p = (g^b)^a \text{ mod } p = A^b \text{ mod } p$  is this uses **Trapdoor functions**, easy to compute but hard to revert.

**Man-In-The-Middle (MITM).** 1. Eve sends  $E_B$  to Alice. Alice computes  $K_{AE}$   
Vanilla DH is Vulnerable to Active Attack, 2. Eve sends  $E_A$ . Bob computes  $K_{BE}$   
DH guarantees key agreement, **not** authenticity. 3. Creates an Alice-Eve channel and Eve-Bob channel  
**Attacker could intercept A from Alice and B she can read, modify, and relay all communication. from Bob, and then impersonate.**

**Asymmetric Cryptography.** Encryption using a public-private key pair. Public key can be stored on a public server, the **Public Key Infrastructure (PKI)**.  
**Hash Functions.** maps any-length message to a fixed-size output using a hash function.  $h(M) = H$   
**Security properties.** 1. **Pre-image resistance.** Given  $H = h(M)$ , it is hard to find M.  
2. **Second pre-image resistance.** Given M, it is hard to find another  $M' \neq M$  such that  $h(M') = h(M)$ .  
3. **Collision resistance.** It is hard to find any two values M and M' such that  $h(M) = h(M')$ .

**Block Cipher - Counter Mode** **Block Cipher - CBC-MAC.** turns block cipher into MAC by chain-add randomness per-block without using message blocks with encryption.  
inter-block dependencies.  $C_0 = IV$  **Encrypt.**  $C_i = E_K(M_i \oplus C_{i-1})$  MAC $_K(M_1, \dots, M_n) = C_n$   
(parallel) deterministic. fixed IV → same message/same MAC.  
**Encrypt.**  $C_i = M_i \oplus E_K(\text{Nonce} + i)$ . △message length must be known and fixed; otherwise, extension  
**Decrypt.**  $M_i = C_i \oplus E_K(\text{Nonce} + i)$ . attacks are possible (e.g., using  $M_i(T \oplus M')$ ).  
Nonce must be unique (never with same key) reuse leaks keystream.  
**1. Encrypt-and-MAC.** send  $E_K(P_1 P_2)$  and  $MAC_K(P_1 P_2)$ . integrity of plaintext but no integrity on ciphertext/may reveal info on ciphertext.  
**2. MAC-then-Encrypt** send  $E_K(P_1 P_2 || MAC_K(P_1 P_2))$ .  
**Message Authentication Code** short tag computed from hides MAC and provides confidentiality/integrity on plaintext but a message and secret key(pre-no integrity ciphertext shared) to verify message  
**3. Encrypt-then-MAC** send  $E_K(P_1 P_2) || MAC_K(E_K(P_1 P_2))$ . **authenticates** ciphertext, ensures **confidentiality/integrity** ✓